교육부 Ministry of Education

NRF 한국연구재단 National Research Foundation of Korea

공유교과목 [딥러닝]

Perceptron

Convergence and Open Sharing System COSS 인공지능

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Perceptron

Perceptron

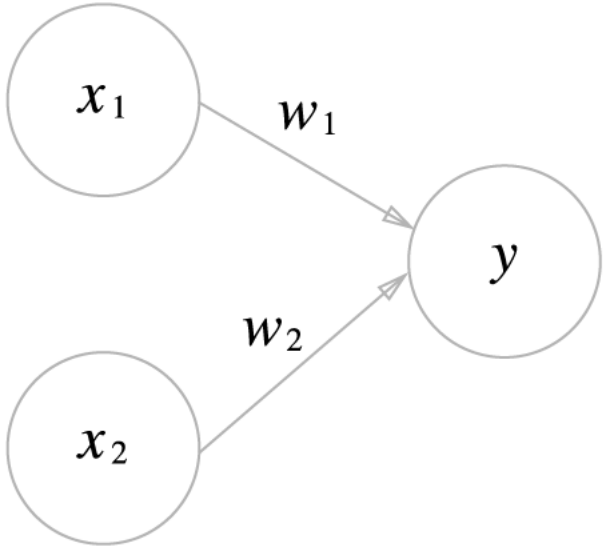
Definition

- An algorithm for learning **a binary classifier** called **a threshold function**: a function that m aps its input **x** (a real-valued vector) to an output value **y** (a single binary value)

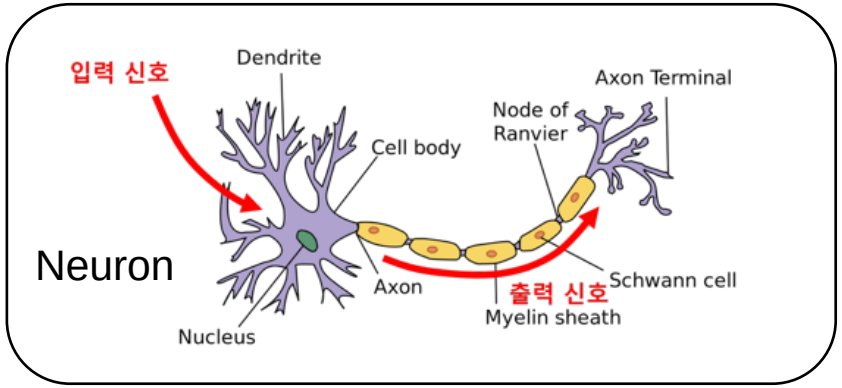
- There are weight **w** (a real-value vector) and threshold $\boldsymbol{\theta}$

0.5

- The circles below are called neuron or node



$$y = \begin{cases} 0 & (w_1 x_1 + w_2 x_2 \leq \theta) \\ 1 & (w_1 x_1 + w_2 x_2 > \theta) \end{cases}$$

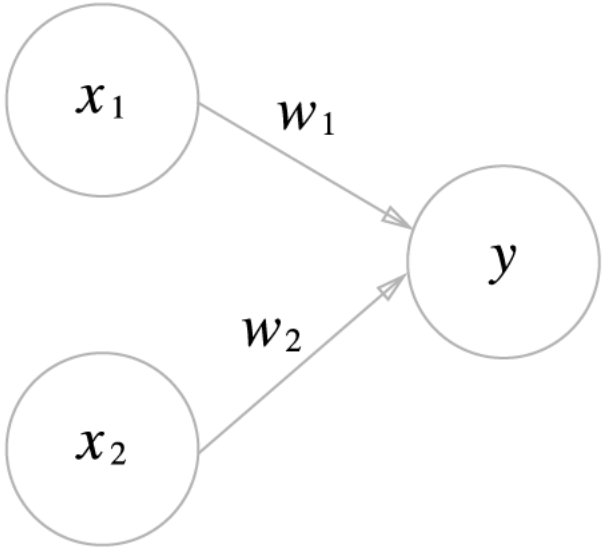


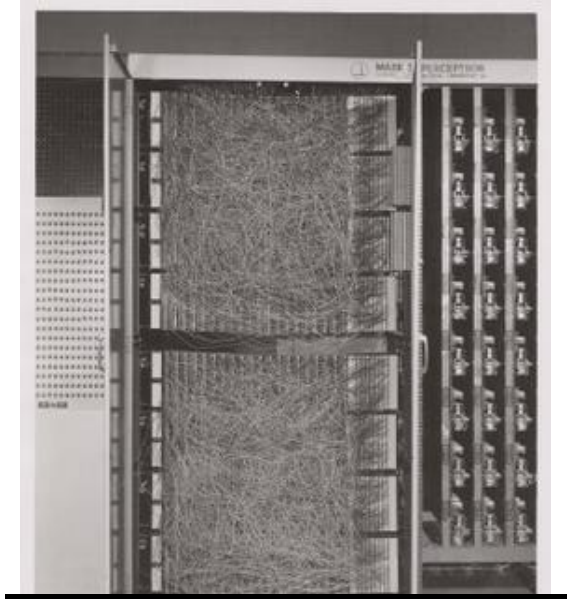
Perceptron

History

- Invented in 1958 at the Cornell Aeronautical Laboratory by Frank Rosenblatt

- Based on Rosenblatt's statements, The New York Times reported the perceptron to be "*the embryo of an electronic computer that [the Navy] expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence*."



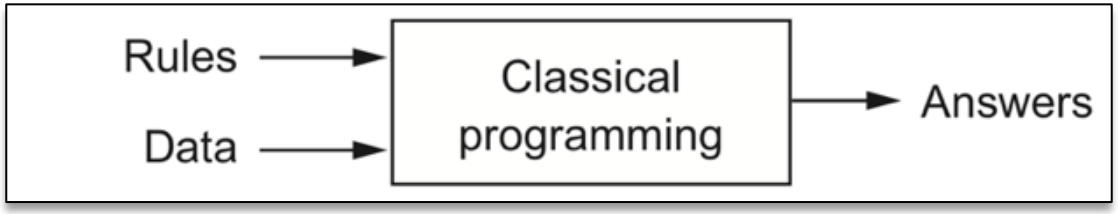


Mark I Perceptron machine

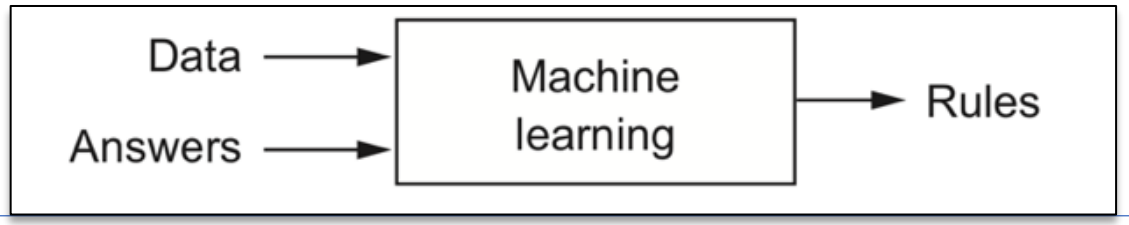
Perceptron

What made the Perceptron so popular?

- **Before the Perceptron**, rule-based programs were difficult because:
  - Complex problems that are difficult to program
  - When a problem occurs in the program, it is difficult to maintain
  - The environment is constantly changing, so the program must be constantly updated



- **After the Perceptron**, we don't need rule-based programs because:
  - Machine learning allows computers to program rules by themselves through learning given input (data) **x** & results (answers) **y**



Perceptron

Types of data

- **Quantitative data** (numerical data): Data whose size has a meaningful numerical value
  - Continuous data: data with continuous values (e.g., height, weight)
  - Discrete data: numerical meaning but not decimal (e.g., number of births)

- **Qualitative data** (categorical data): data divided into several categories
  - Ordinal data: there is an ordinal relationship (e.g., stage of disease, grade)
  - Nominal data: simply classified data (e.g., gender, blood type)

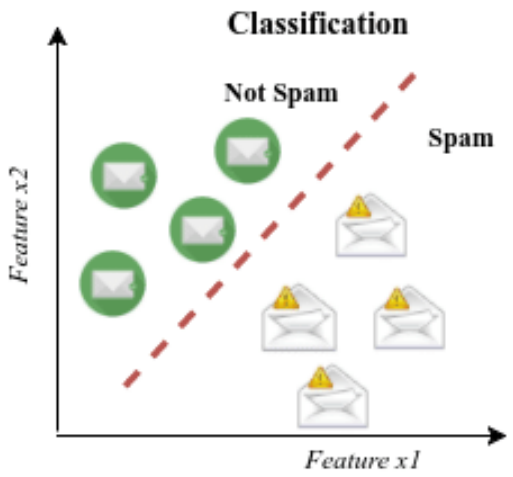
Types of data			**Examples**
Quantitative data (numerical data)	Continuous data	Ratio scale	Height; Weight
		Interval scale	Temperature; Stock price
	Discrete data		Number of birth
Qualitative data (categorical data)	Ordinal data		Stage of disease; Grade
	Nominal data		Gender; Blood type

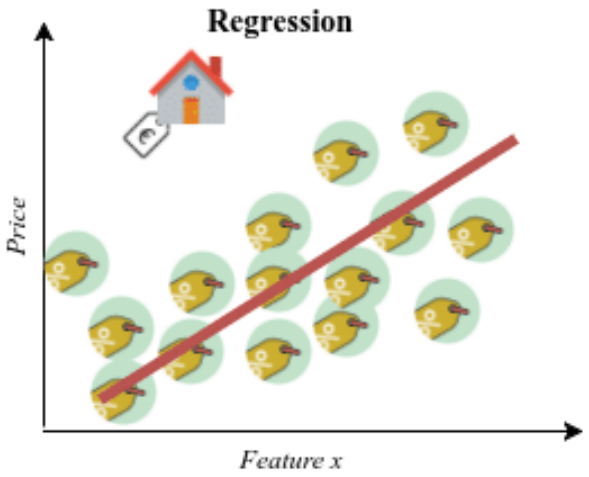
Perceptron

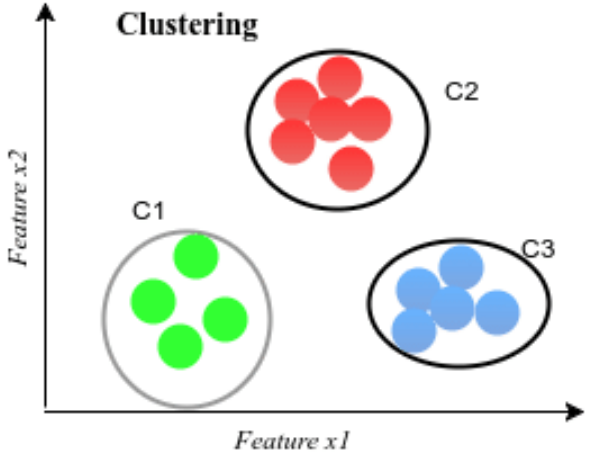
Types of machine learning

- **Supervised learning**: learning by giving input and output (result, answer) values
  - Classification: If the output is categorical data (e.g., spam classification)
  - Regression: When the output is continuous data (e.g., house price prediction)

- **Unsupervised learning**: learning only by input
  - Clustering: The computer groups data with similar characteristics based on the input (e.g., user classification)





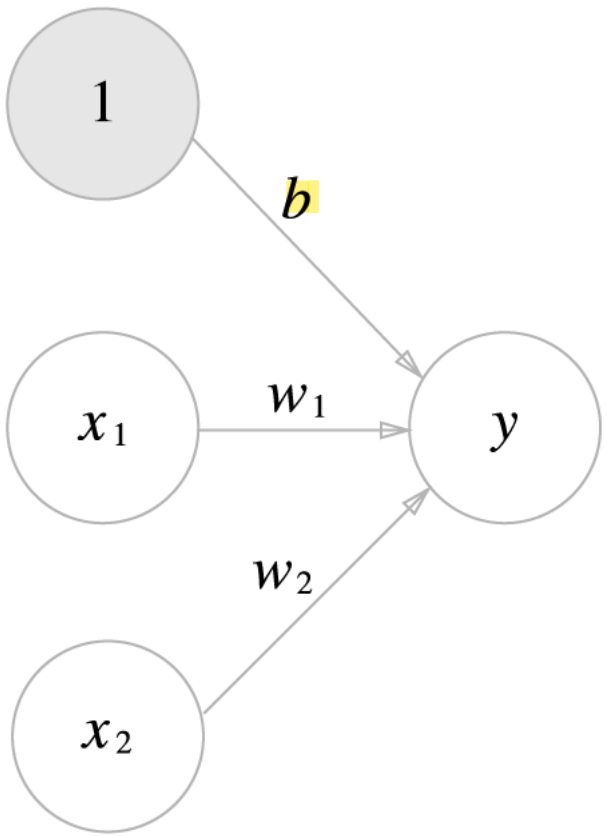


ref: DEEP GENERATIVE MODELS FOR SYNTHETIC RETINAL IMAGE GENERATION

Perceptron

Introduction of bias

- Binary classification centered on 0 by replacing ***θ***(theta) with **-b**(bias)



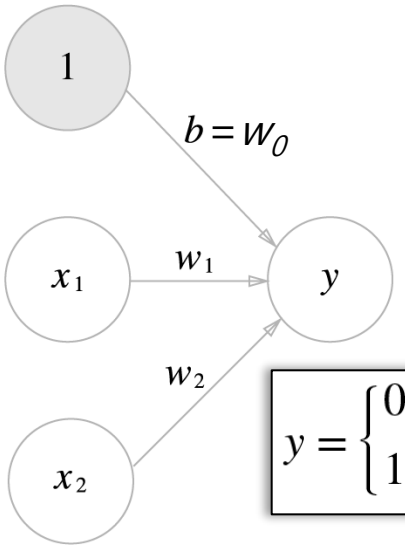
$$y = \begin{cases} 0 & (w_1x_1 + w_2x_2 \le \theta) \\ 1 & (w_1x_1 + w_2x_2 > \theta) \end{cases}$$

$$\boxed{y = \begin{cases} 0 & (b + w_1x_1 + w_2x_2 \le 0) \\ 1 & (b + w_1x_1 + w_2x_2 > 0) \end{cases}}$$

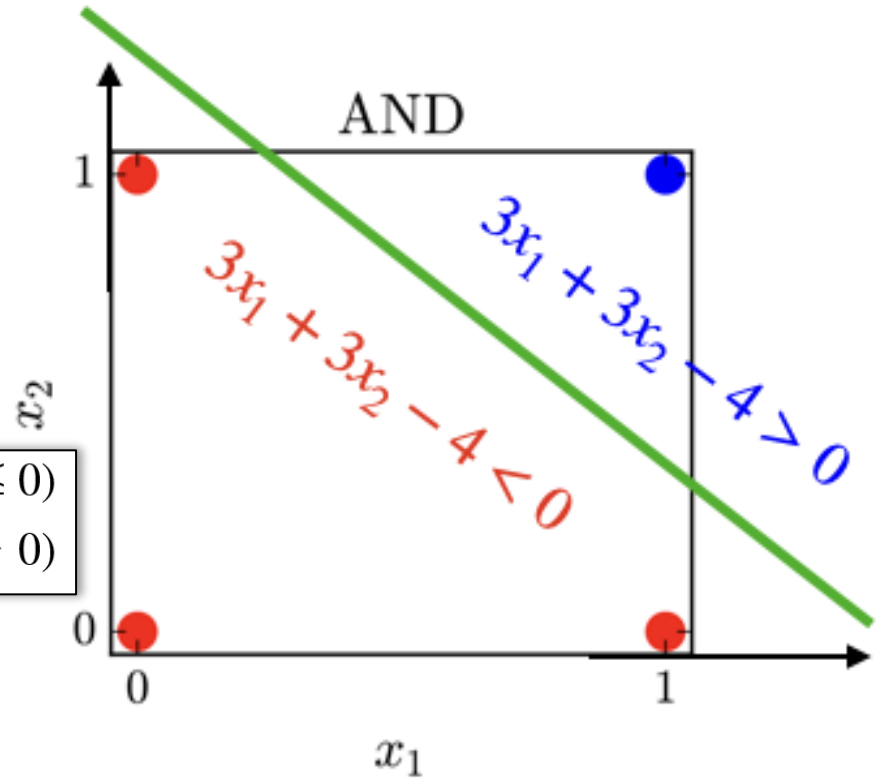
Single-Layer Perceptron – AND gate

AND

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1



$$y = \begin{cases} 0 & (b + w_1x_1 + w_2x_2 \leq 0) \\ 1 & (b + w_1x_1 + w_2x_2 > 0) \end{cases}$$



$$(w_0, w_1, w_2) = (-4,3,3)$$

$$\hat{y} = \sum_0^2 w_i x_i = -4 + 3x_1 + 3x_2$$

$-4 + 3 \times 0 + 3 \times 0 = -4$ $\quad \hat{y} = 0$

$-4 + 3 \times 0 + 3 \times 1 = -1$ $\quad \hat{y} = 0$

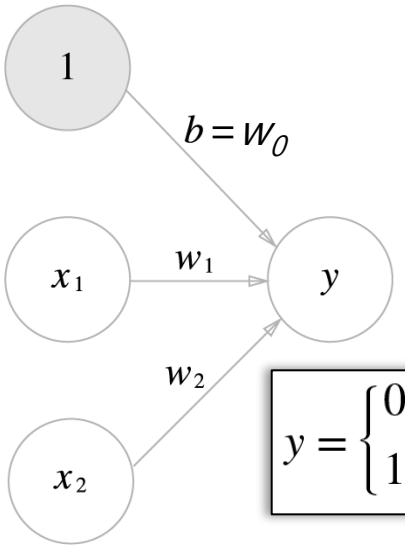
$-4 + 3 \times 1 + 3 \times 0 = -1$ $\quad \hat{y} = 0$

$-4 + 3 \times 1 + 3 \times 1 = 2$ $\quad \hat{y} = 1$

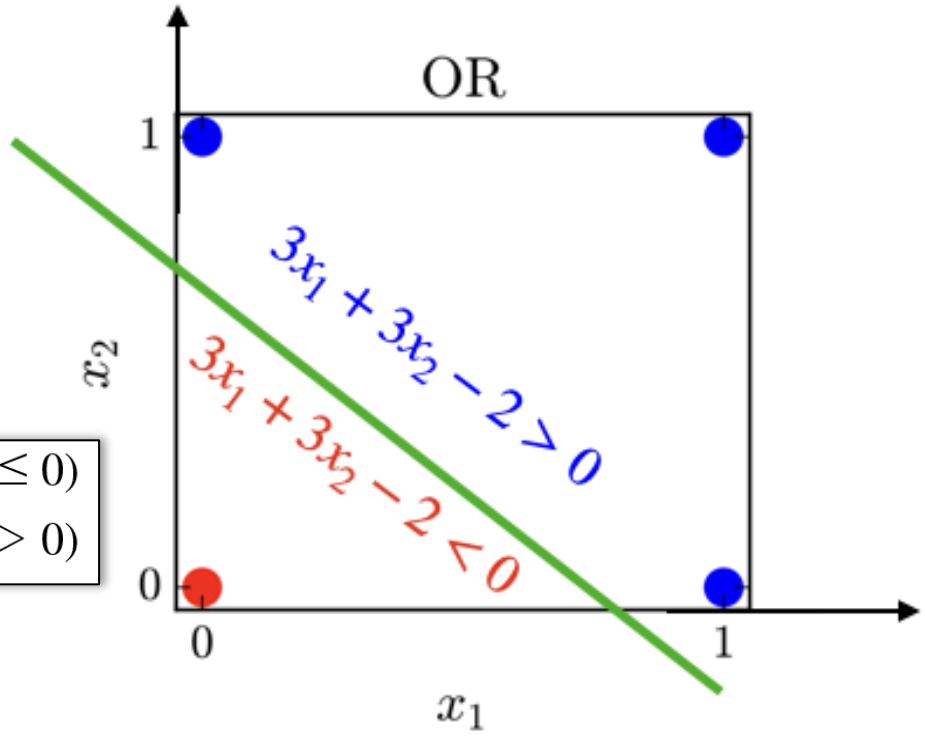
Single-Layer Perceptron – OR gate

OR

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1



$$y = \begin{cases} 0 & (b + w_1x_1 + w_2x_2 \le 0) \\ 1 & (b + w_1x_1 + w_2x_2 > 0) \end{cases}$$



$$(w_0, w_1, w_2) = (-2,3,3)$$

$$\hat{y} = \sum_0^2 w_i x_i = -2 + 3x_1 + 3x_2$$

$-2 + 3 \times 0 + 3 \times 0 = -2$ $\quad$ $\hat{y} = 0$

$-2 + 3 \times 0 + 3 \times 1 = 1$ $\quad$ $\hat{y} = 1$

$-2 + 3 \times 1 + 3 \times 0 = 1$ $\quad$ $\hat{y} = 1$

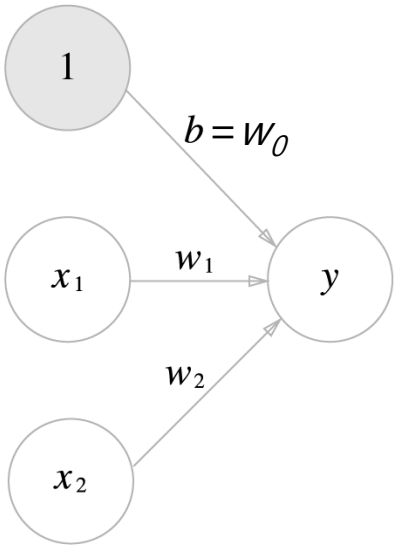
$-2 + 3 \times 1 + 3 \times 1 = 4$ $\quad$ $\hat{y} = 1$

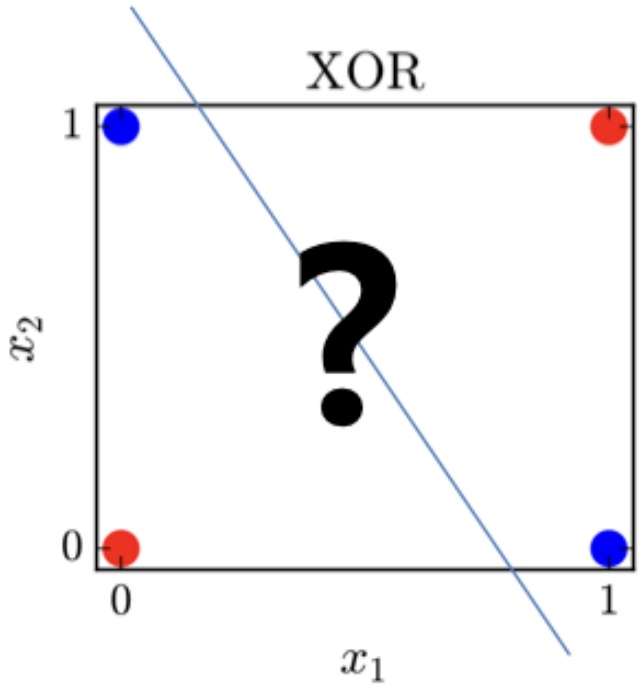
Limitations of the Single-Layer Perceptron

단층 퍼셉트론에 한계

XOR

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0





- "*Perceptrons: an introduction to computational geometry*" (Marvin Minsky & Seymour Papert, 1969)

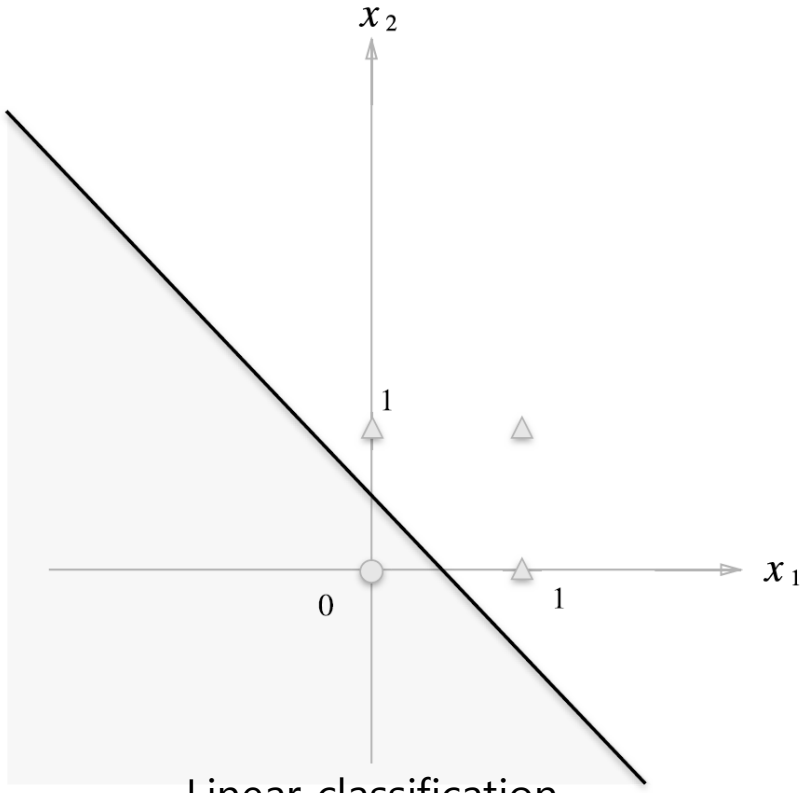
- The single-layer perceptron is a linear classification method and therefore, has limitations

Limitations of the Single-Layer Perceptron

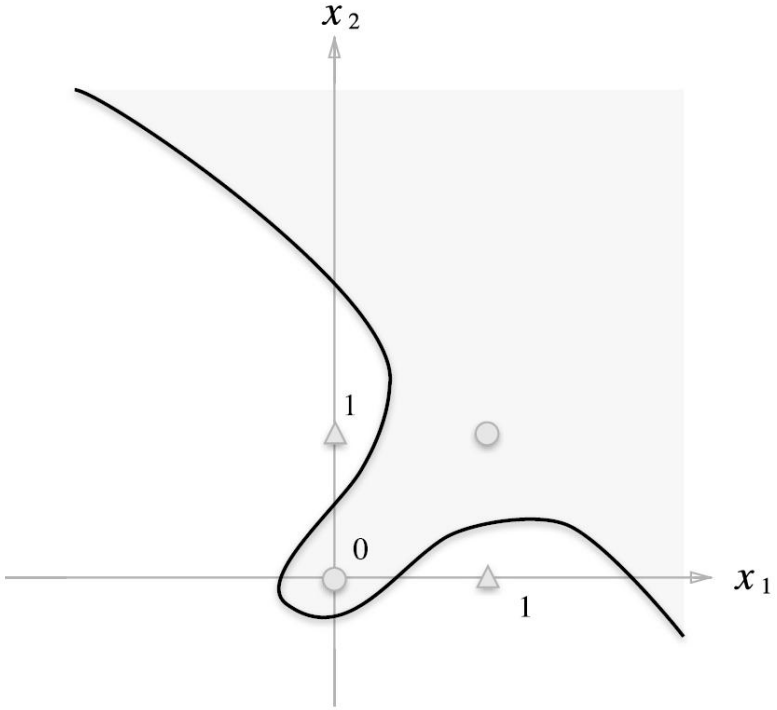
Linear vs. Non-linear classification

- **Linear classification**: An arbitrary line can classify the two classes

- **Non-linear classification**: To classify the two classes, we need non-linearity



Linear classification

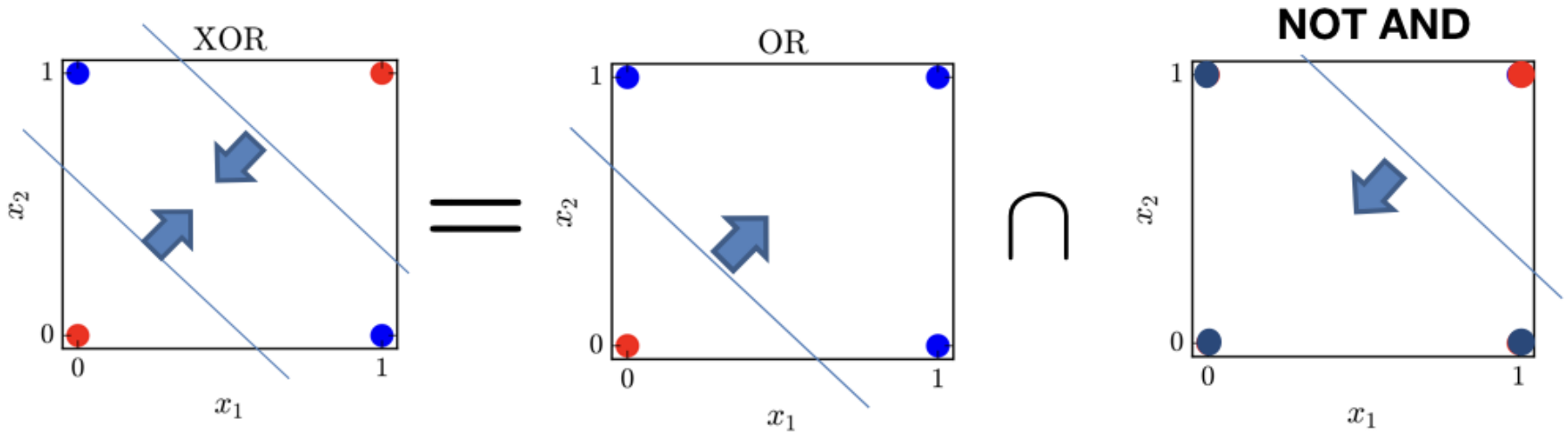


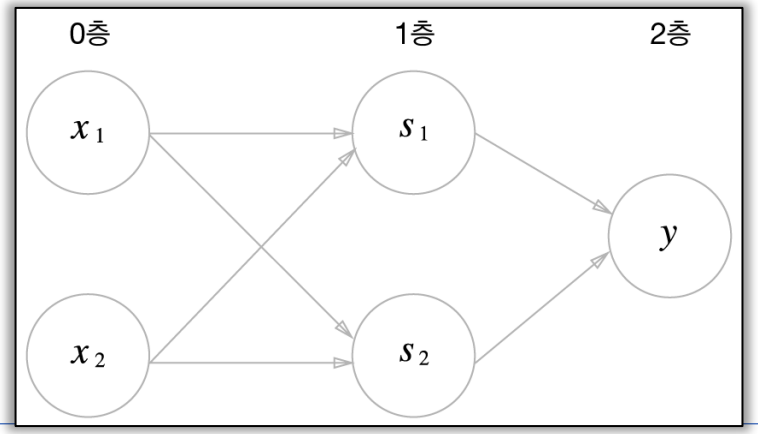
Non-linear classification

Limitations of the Single-Layer Perceptron

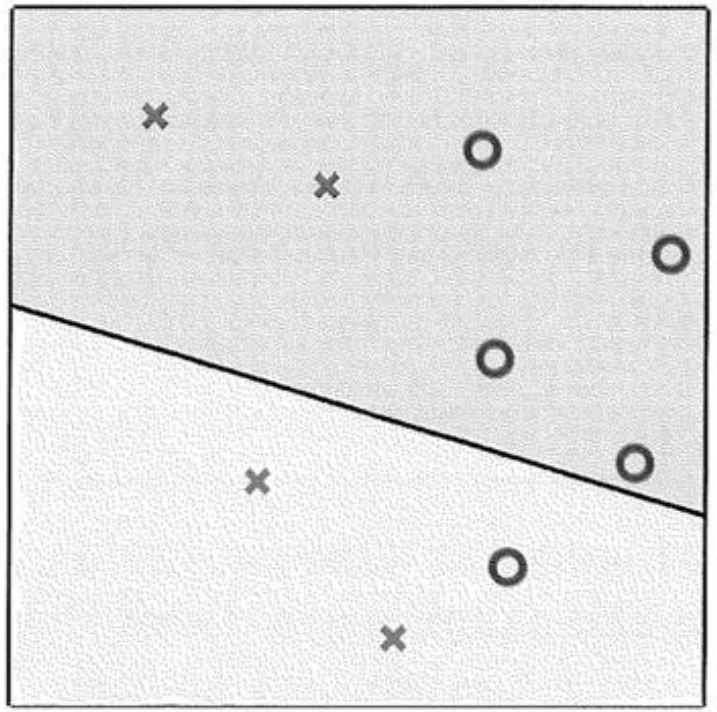
So, how to solve XOR gate problem?

- Use multiple gates at the same time → use **multi-layer perceptron**

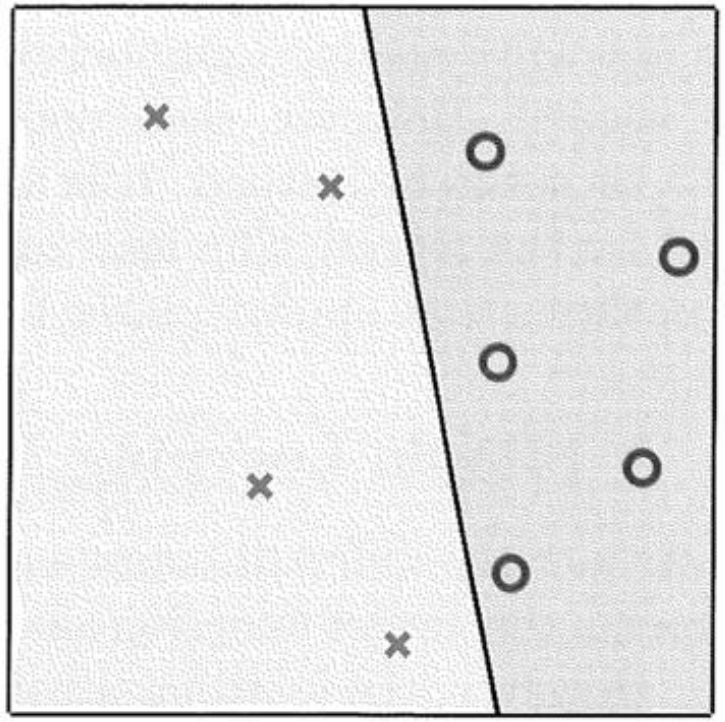




Supervised learning using "Perceptron"



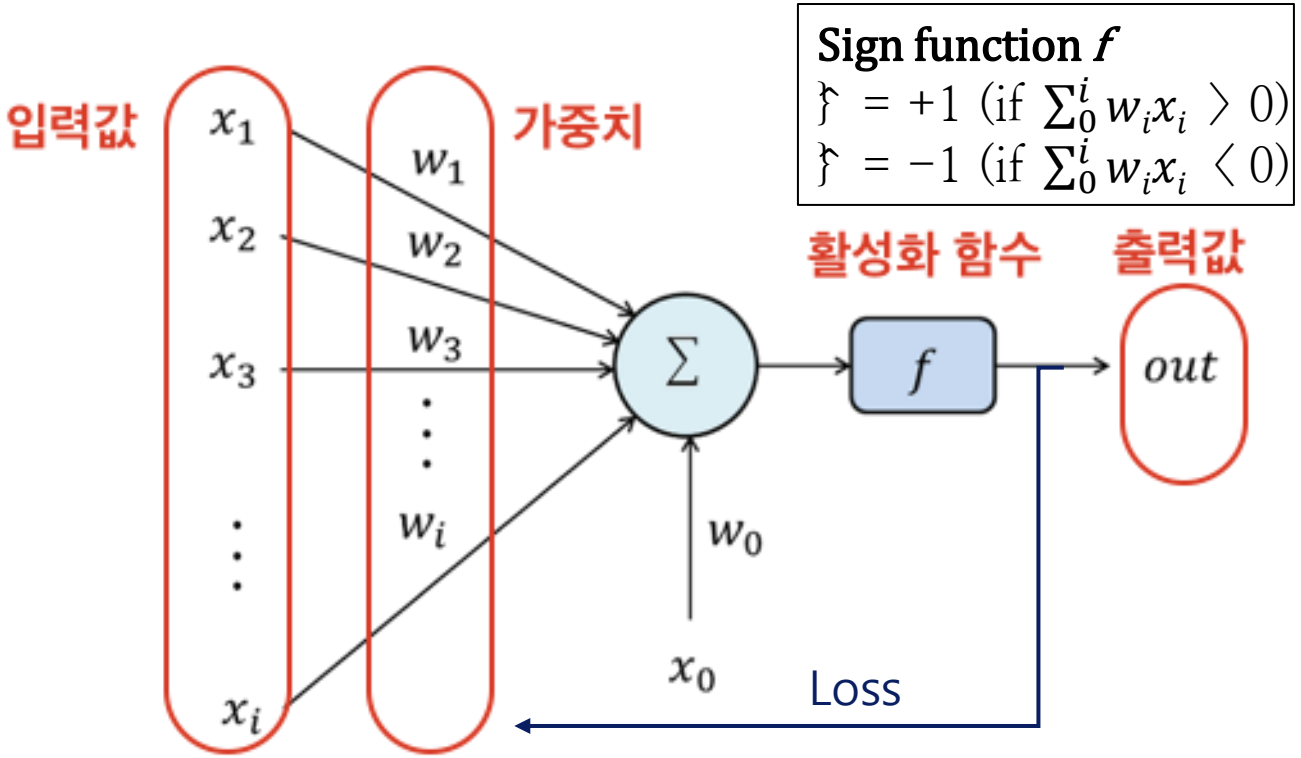
Incorrect linear classification



Correct linear classification

- Using a perceptron, how to make a computer automatically do a linear classification?

Supervised learning using "Perceptron"



Sign function f

$f = +1$ (if $\sum_0^i w_i x_i > 0$)

$f = -1$ (if $\sum_0^i w_i x_i < 0$)

- Loss = real value – out value
  = y - $\hat{y}$
  = y - **sign**$[(\sum_0^i w_i x_i)]$

- Learning rate η(eta)

- **Supervised learning** (example)

$\mathbf{w}_{new} = \mathbf{w}_{old} + \eta(y - \hat{y})\mathbf{x}$

$= \mathbf{w}_{old} + \eta(y - \mathbf{sign}[(\sum_0^i w_i x_i)])\mathbf{x}$

- It is important to set an appropriate learning rate!

Supervised learning using "Perceptron"

Supervised learning example

- Apply stochastic gradient-descent method

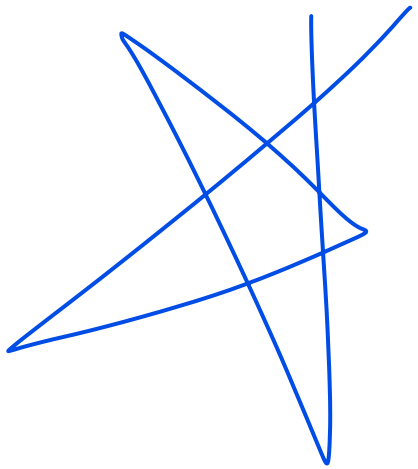
1) Arbitrarily set weight w and learning rate η

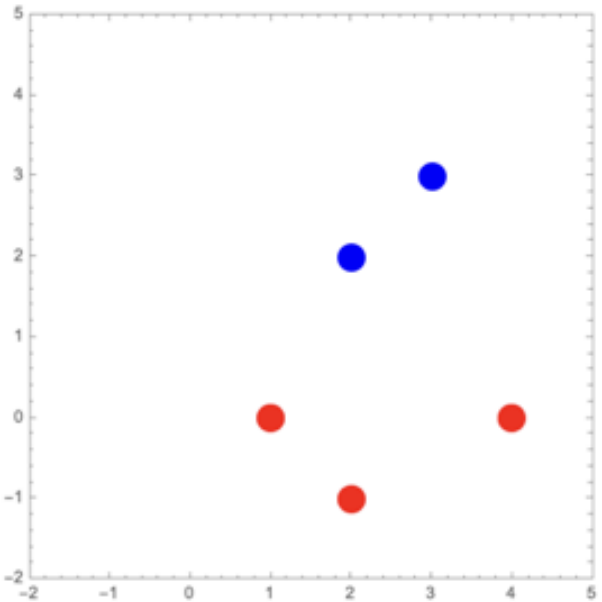
2) Pick a random x misclassified by the perceptron

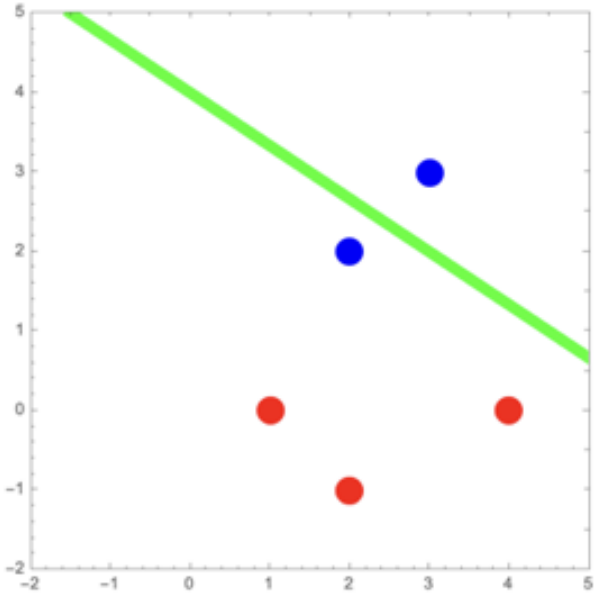
3) Update weight w using "$\mathbf{w}_{new} = \mathbf{w}_{old} + \eta(y - \hat{y})\mathbf{x}$"

4) Repeat steps 2-3

Supervised learning using "Perceptron" (example)







There are 5 **x**, **y** data

x = $[x_0\ x_1\ x_2]$ = $[1\ x_1\ x_2]$

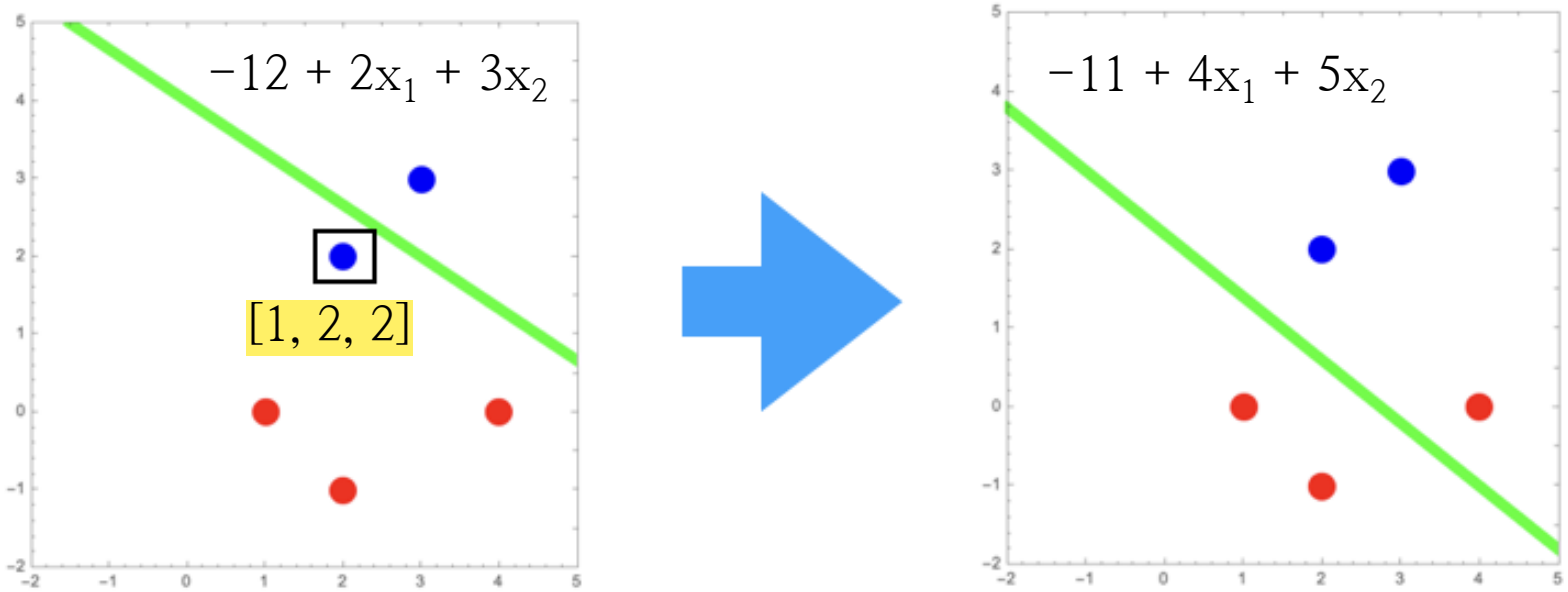
y = **1** or **-1**

w = $[w_0\ w_1\ w_2]$ = [-12, 2, 3]

→ **wx** = $w_0x_0 + w_1x_1 + w_2x_2 = -12x_0 + 2x_1 + 3x_2$

$\boldsymbol{\eta}$ = 0.5

Supervised learning using "Perceptron" (example)



Dissatisfied with the classification $\mathbf{x}$ = [1, 2, 2], y = **+1**, $\hat{y}$ = **-1**

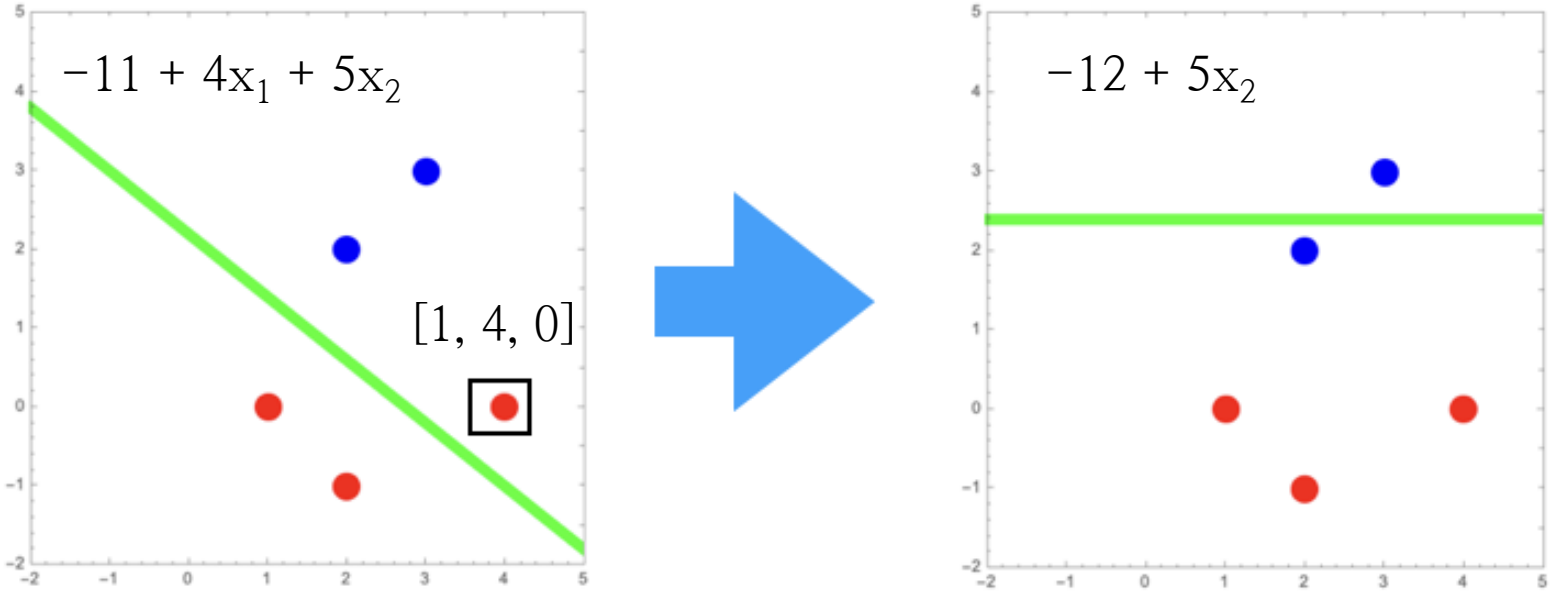
$\mathbf{w}_{new} = \mathbf{w}_{old} + \eta(y - \hat{y})\mathbf{x}$ update!

$= [-12, 2, 3] + 0.5(+1 - (-1))[1, 2, 2]$

$= [-12, 2, 3] + [1, 2, 2] = [-11, 4, 5]$

$\mathbf{w}_{new}\mathbf{x} = w_0x_0 + w_1x_1 + w_2x_2 = -11 + 4x_1 + 5x_2$

Supervised learning using "Perceptron" (example)



Dissatisfied with the classification $\mathbf{x}$ = [1, 4, 0], y = **-1**, $\hat{y}$ = **+1**

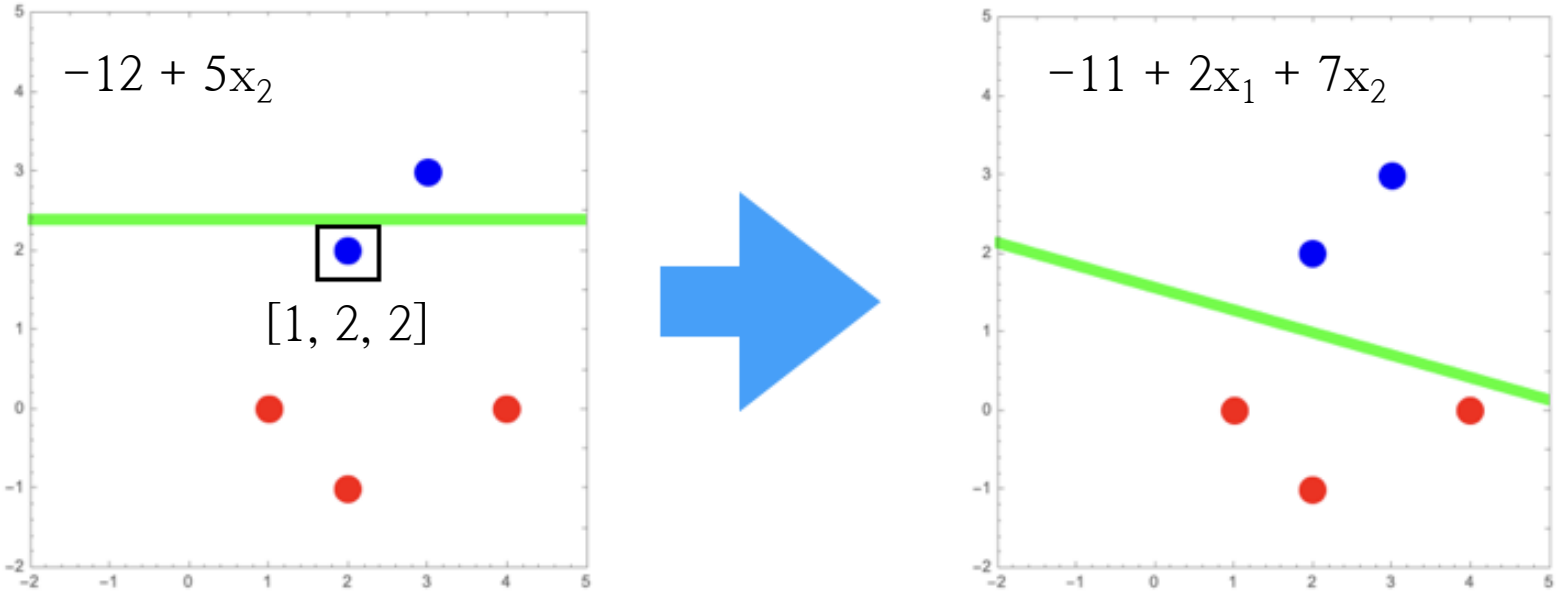
$\mathbf{w}_{new} = \mathbf{w}_{old} + \eta(y - \hat{y})\mathbf{x}$ update!

$= [-11, 4, 5] + 0.5(-1 - (+1))[1, 4, 0]$

$= [-11, 4, 5] - [1, 4, 0] = [-12, 0, 5]$

$\mathbf{w}_{new}\mathbf{x} = w_0x_0 + w_1x_1 + w_2x_2 = -12 + 5x_2$

Supervised learning using "Perceptron" (example)



Dissatisfied with the classification $\mathbf{x}$ = [1, 2, 2], y = **+1**, $\hat{y}$ = **-1**

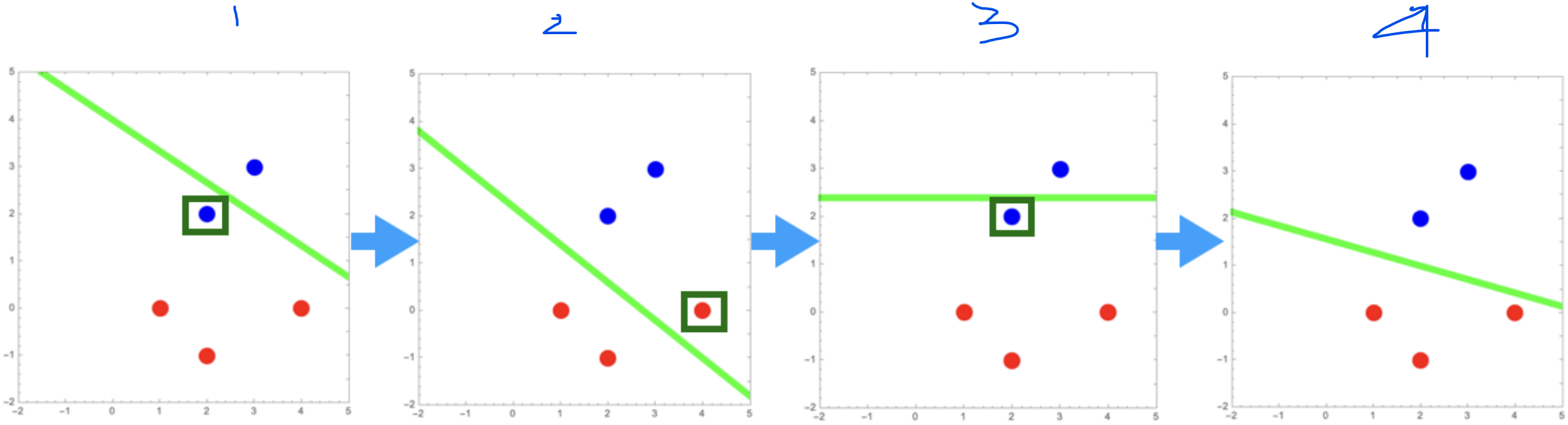
$\mathbf{w}_{new} = \mathbf{w}_{old} + \eta(y - \hat{y})\mathbf{x}$ update!

$= [-12, 0, 5] + 0.5(1 - (-1))[1, 2, 2]$

$= [-12, 0, 5] + [1, 2, 2] = [-11, 2, 7]$

$\mathbf{w}_{new}\mathbf{x} = w_0x_0 + w_1x_1 + w_2x_2 = -11 + 2x_1 + 7x_2$

Supervised learning using "Perceptron" (example)



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3) Update weight w using "$\mathbf{w}_{new} = \mathbf{w}_{old} + \eta(y - \hat{y})\mathbf{x}$"
4) Repeat steps 2-3

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thank you

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디지털신기술인재양성 혁신공유대학사업의 연구결과입니다.

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